Foundations of Programming

Assignment No.: 18

PROBLEM STATEMENT: Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check of two strings iv) Check palindrome v) Check substring.

OBJECTIVE:

- To understand basic string operations in C.
- To learn how to manipulate strings using loops and functions.
- To perform comparison, reversal, and substring checking.
- To check whether a string is a palindrome.

THEORY:
In C programming, a string is a sequence of characters terminated by a null character ('\0'). Strings are commonly stored in character arrays. Various operations can be performed on strings such as calculating their length, reversing the characters, comparing two strings, checking whether a string is a palindrome, and finding a substring within another string. These operations can be implemented using loops and conditional statements or by using standard library functions from the string.h header file.

PLATFORM: Linux
CODE:

```
#include <stdio.h>
#include <string.h>
int main()
{
   char str1[100],str2[100],rev[100];
   int i,len;
   printf("Enter first string: ");
   scanf("%s",str1);
   printf("Enter second string: ");
   scanf("%s",str2);
```

```
    len=strlen(str1);
    printf("Length of string = %d\n",len);
    for(i=0;i<len;i++)
        rev[i]=str1[len-i-1];
    rev[i]='\0';
    printf("Reversed string = %s\n",rev);
    if(strcmp(str1,rev)==0)
        printf("The string is Palindrome\n");
    else
        printf("The string is not Palindrome\n");
    if(strcmp(str1,str2)==0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");
    if(strstr(str1,str2)!=NULL)
        printf("Substring found in main string\n");
    else
        printf("Substring not found\n");
    return 0;
}
```

INPUT: A string and another string for comparison and substring checking.

```
Enter first string: radar
Enter second string: monsoon
```

OUTPUT: Displays string length, reversed string, equality result, palindrome result, and substring check result.

```
Enter first string: radar
Enter second string: monsoon
Length of string = 5
Reversed string = radar
The string is Palindrome
Strings are not equal
Substring not found
```

SAMPLE INPUT:

Enter a string: madam
Enter another string for equality check: mad

SAMPLE OUTPUT:

Length of string = 5

Reversed string = madam

The string is a Palindrome

Strings are not equal

Substring found in main string

```c
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[100],str2[100],rev[100];
    int i,len;
    printf("Enter first string: ");
    scanf("%s",str1);
    printf("Enter second string: ");
    scanf("%s",str2);
    len=strlen(str1);
    printf("Length of string = %d\n",len);
    for(i=0;i<len;i++)
        rev[i]=str1[len-i-1];
    rev[i]='\0';
    printf("Reversed string = %s\n",rev);
    if(strcmp(str1,rev)==0)
        printf("The string is Palindrome\n");
    else
        printf("The string is not Palindrome\n");
    if(strcmp(str1,str2)==0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");
    if(strstr(str1,str2)!=NULL)
        printf("Substring found in main string\n");
    else
        printf("Substring not found\n");
    return 0;
}
```

CONCLUSION: Thus, the C program successfully performs different string operations such as length calculation, reversal, equality check, palindrome check, and substring detection.

FAQs:

1. What is a string in C programming?
2. Which header file is required for string functions?
3. What function is used to find the length of a string?
4. What is a palindrome string?
5. Which function is used to check a substring in C?